The Frohmanian Tether on SDiff($\mathbb{T}^3$): A Metriplectic Correction, Its Canonicity, and an Obstruction to Damping

Benjamin Stanley Frohman*
Independent Researcher
ORCID: 0009-0006-7068-3718 frohmanbenjamin@gmail.com

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Abstract

We present in full both (1) the complete original geometric and physical framework of the **Frohmanian Symplectic Tether** $\mathfrak{T}_F$ (preserving without omission all original definitions, physical motivations, conjectured geometric–analytic bridges, and extensions from *Frohmanian_Tether_Final_Submission_v8*), and (2) its complete, kernel-verified, non-vacuous conditional formalization in Lean 4 (repository `NS_Millennium_Proof`: **537 theorems across 23 modules, zero `sorryAx`, and only the standard axioms `propext`, `Classical.choice`, `Quot.sound`**).

Let $M(t) = \|\omega(t)\|_{L^\infty}$ denote the vorticity sup-norm (`vorticity_sup_norm`) along a physical solution (`IsPhysicalNSSolution`) of the three-dimensional incompressible Navier–Stokes equations with smooth, divergence-free, finite-energy initial data and viscosity $\nu > 0$. We prove in Lean 4 (`frohmanian_tether_theorem` and `conditional_regularity_of_open_hyps`) that if (i) Kato local existence holds with a uniform restart length on the class of physical solutions (`IsPhysicalNSSolution`, enforcing pointwise time-differentiability and the Leray L^2 energy inequality), (ii) uniqueness holds within `IsPhysicalNSSolution`, and (iii) $M(t)$ satisfies the Frohmanian cubic differential inequality (`FrohmanianTetherRate`)

$$M'(t) \le C_{\mathrm{CZ}}\, M(t)^2 - \kappa''\, M(t)^3, \qquad t \ge 0,$$

for constants $C_{\mathrm{CZ}}, \kappa'' > 0$, then a scalar comparison argument (`comparison_ode_stability`) yields the uniform ceiling $M(t) \le \max\bigl(M(0), C_{\mathrm{CZ}}/\kappa''\bigr)$ (`FrohmanianTetherBound`), and, as a Lean-checked implication from these open hypotheses, the Beale–Kato–Majda criterion together with uniform Kato restart yields a smooth, divergence-free physical solution.

Crucially, the time-smoothness and physical-energy repair (`IsPhysicalNSSolution`) is **fully implemented and kernel-verified non-vacuous**: in `ClayStatements.lean`, we prove that unphysical time-step fields are excluded (`stepField_not_physical`) and that all hypotheses of `conditional_regularity_of_open_hyps` (`repaired_hInt`, `repaired_huniq_at_zero`, `frohmanianTetherBound_zero`, `frohmanianTetherRate_zero`) hold simultaneously and unconditionally at $u_0 = 0$ (`conditional_regularity_nonvacuous_at_zero`). Furthermore, every original conjectured bridge including `reduced_pairing_of_unmodified_and_tether` (via `ConjecturedUnmodifiedTetherPairing`), `tethered_stretching_of_weight`, `tether_coadjoint_invariance_of_integral_eq` (via `ConjecturedSDiffPushforwardInvariance`), and the symmetric uniqueness characterization (`uniqueness_of_minimal_tether`, which delegates to `uniqueness_of_minimal_symmetric_tether`)is compiled in the Lean build with zero `sorry`s.

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Status of the damping premise. The full tethered bracket $\{F,G\}_{\mathfrak{T}_F} = \{F,G\}_{\mathrm{cl}} + T(F,G)$ is of *metriplectic type*: an antisymmetric Lie–Poisson part plus a symmetric (`tetherKernel_symmetric`), negative-semidefinite (`tetherKernel_diagonal_nonpos`) tether T, with the classical part antisymmetric (`classicalBracket_antisymm`). The Jacobi identity of the classical part is cited from [1], not formalized. It is *not* antisymmetric and no Jacobi identity is claimed for it. Because T is symmetric and degenerate on the kinetic energy (C2), $T(M,H) = 0$ for every functional M (`tetherKernel_left_H_zero`), so the bracket contributes *no* damping to the evolution of any functional, including $M(t) = \|\omega\|_{L^\infty}$: the damping coefficient derived from the bracket is 0. The cubic rate (R) and the resulting closing bound are therefore **open assumptions**, not consequences of the tether. Every statement below that passes through (R)—the Riccati/cubic comparison ceiling and the conditional regularity theorems—depends on this unresolved premise. **No regularity claim for the Navier–Stokes equations is made.**

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1 Introduction

The incompressible Navier–Stokes equations on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$ (or on $\mathbb{R}^3$ with finite energy) are (Lean: `NS_PDE`)

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0, \qquad u(0) = u_0, \qquad (\texttt{NS_PDE}) \tag{1}$$

with $\nu > 0$. Taking the curl yields (`vorticity_transport_at`, under the regularity hypothesis `VorticityTransportRegularity`) the vorticity transport equation for $\omega = \nabla \times u$:

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega. \tag{2}$$

By the Beale–Kato–Majda theorem [3] (cited, not formalized), a smooth solution can be continued past T whenever $\int_0^T \|\omega(t)\|_{L^\infty}\, dt < \infty$. A uniform ceiling on $M(t) = \|\omega(t)\|_{L^\infty}$ therefore implies, by BKM (cited, not formalized), smooth continuation of that solution. Obtaining such a bound for arbitrary large smooth data is the core difficulty of the Clay Millennium Problem [5] (background, not formalized): the vortex-stretching term $(\omega \cdot \nabla)u$ is supercritical with respect to the natural parabolic scaling

$$u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t), \qquad \omega_\lambda(x,t) = \lambda^2 \omega(\lambda x, \lambda^2 t), \qquad (\text{at each fixed time: } \texttt{ScalingIdentities.vorticity_} \tag{3}$$

which leaves (1) invariant at fixed ν.

The Frohmanian programme approaches this problem through a two-layer architecture:

1. **Geometric layer (`ArnoldGeometric.lean`, `SymplecticTether.lean`, `Uniqueness.lean`, `ForMathlib/Projection.lean`):** Starting from Arnold's coadjoint orbit of volume-preserving diffeomorphisms [1, 2], we introduce the **Frohmanian Symplectic Tether** $\mathfrak{T}_F$realized concretely via the L^2-projected quadratic kernel `TetherKernel` with universal Calderón–Zygmund coupling $\kappa = C_{\mathrm{CZ}}(3) = 3/2$. By projecting functional derivatives orthogonally to the velocity field $u = \delta H/\delta\omega$ via the Gram–Schmidt projector Π_u (`Pi_u`), the tether satisfies exact degeneracy $B(F,H) = 0$ on the kinetic-energy Hamiltonian H (`tetherKernel_degenerates_on_kinetic_energy`), leaving the reversible Euler vector field unmodified while supplying a negative-semidefinite diagonal value $T(F,F) = -\kappa\|\omega\|_{L^2}^2 \int |\Pi_u \delta F/\delta\omega|^2$ (`ProducesControllableNegativeFeedback`). The same degeneracy implies that the tether induces no damping of $M(t)$ (Section 5.2; `tetherKernel_left_H_zero`).

2. **Analytic layer (`NS_Equations.lean`, `TetheredLyapunov.lean`, `IndependentMajorant.lean`, `CubicSourceCertificate.lean`, `AnalyticPipeline.lean`, `ClosureBlueprint.lean`, `ConditionalRegularity.lean`, `ClayStatements.lean`):** Using the quartic Lyapunov functional $S_\varepsilon(t)$ weighted by the transported enstrophy-accumulation scalar φ (`LyapunovS`, `IsEnstrophyAccumulation`), the programme links the cubic rate $M'(t) \le C_{\mathrm{CZ}} M(t)^2 - \kappa'' M(t)^3$ (`FrohmanianTetherRate`)—an *open assumption* not derived from the tether (Section 5.2)—to the autonomous comparison ODE $y' = Cy^2 - \kappa'' y^3$ (`comparison_ode_stability`, `comparison_majorant_global_bound`), yielding the uniform ceiling $M(t) \le \max(M(0), C_{\mathrm{CZ}}/\kappa'')$ (`riccati_ceiling_of_vorticity_di`) and, *conditionally on that assumption and the further open hypotheses (K), (U)*, the smooth-continuation conclusion of Theorem 1.1.

Theorem 1.1 (Conditional regularity implication from open hypotheses; `frohmanian_tether_theorem`, `conditional_regularity_of_open_hyps`; depends on the unresolved premise (R))**.** *Let $\nu > 0$ and let u_0 be smooth (`ContDiff`), divergence-free, and of finite L^2 energy. Assume within the class of physical solutions (`IsPhysicalNSSolution`):*

(K) Kato local existence with uniform restart*: a local smooth physical solution (`KatoLocalWitness`) exists from every such datum, with a restart increment $\tau > 0$ extending existence times below $T^*(u_0, \nu)$;*

(R) Frohmanian cubic tether rate (`FrohmanianTetherRate`)*: along every physical solution from u_0, the vorticity sup-norm $t \mapsto M(t)$ is continuous, differentiable on $[0, \infty)$, and satisfies*

$$M'(t) \le C_{\mathrm{CZ}}\, M(t)^2 - \kappa''\, M(t)^3 \qquad (t \ge 0)$$

for constants $C_{\mathrm{CZ}} > 0$, $\kappa'' > 0$ (yielding `FrohmanianTetherBound`);

(U) Physical uniqueness (`huniq`)*: smooth physical solutions (`IsPhysicalNSSolution`) starting from u_0 agree with the Kato witness on each existence interval $[0, T]$.*

Then $M(t) \le \max\bigl(M(0), C_{\mathrm{CZ}}/\kappa''\bigr)$ for all $t \ge 0$, and (1) admits a physical solution defined u on $[0, \infty)$ that is spatially C^∞ for every $t \ge 0$, divergence-free, and unique among all smooth physical solutions.

Verification status of Theorem 1.1.

- **Axiom-clean & sorry-free:** The entire repository `NS_Millennium_Proof` builds cleanly (`lake build NS_Millennium_Proof`, 2717 jobs); its default target contains **537 theorems across 23 modules, zero `sorryAx`, and zero nonstandard axioms** (counted by `collectAxioms` over every theorem; `lake env lean scripts/AxiomAudit.lean` also passes) (depending only on `propext`, `Classical.choice`, `Quot.sound`).
- **Kernel-verified non-vacuous:** The physical-solution predicate `IsPhysicalNSSolution` (Section 4.3) rules out pathological time-step fields that previously exploited Mathlib's totalized derivative `deriv` $= 0$ at jump discontinuities (`stepField_not_physical`). In `ClayStatements.lean`, all hypotheses of Theorem 1.1 are proved to be jointly satisfiable and unconditionally realized at $u_0 = 0$ (`conditional_regularity_nonvacuous_at_zero`).
- **Clear separation of proved implications vs. open premises:** The implication (K) + (R) + (U) $\Longrightarrow$ (the conclusion of Theorem 1.1) is kernel-verified *as an implication* (`Combined.combined_conditional_capstone`). Hypothesis (R) is an open assumption: the tether does not supply it (its damping coefficient is 0, Section 5.2), and no derivation of (R) from (2) is known. Theorem 1.1 therefore depends on an unresolved premise and is *not* a regularity result for Navier–Stokes. Non-vacuity at $u_0 = 0$ shows only that the hypotheses are jointly satisfiable in a trivial case (`conditional_regularity_nonvacuous_at_zero`).

2 Setting and the Physical Solution Class

Write $M(t) = \|\omega(t)\|_{L^\infty}$ (Lean: `vorticity_sup_norm(vorticity(ut))`). In the Lean formalization, the pointwise Navier–Stokes equation is `NS_PDEup`. Because Mathlib's `deriv` returns 0 at points where a function of time is not differentiable, a spatially smooth field with jump discontinuities in time would trivially satisfy `NS_PDE` unless time-differentiability and energy control are enforced.

We therefore define the physical solution class `IsPhysicalNSSolution` in `NS_Equations.lean`:

```
-- NS_Millennium_Proof/Modules/NS_Equations.lean
@[expose] public def IsPhysicalNSSolution (u : TimeDependentVelocity) : Prop :=
  (∀ t ≥ (0 : ℝ), ∀ x : T3, DifferentiableAt ℝ (fun s => u s x) t) ∧
```

```
(∀ t ≥ (0 : ℝ), Continuous (u t)) ∧
(∀ t ≥ (0 : ℝ), Integrable (fun x : T3 => ‖u t x‖ ^ 2)) ∧
(∀ t ≥ (0 : ℝ), (∫ x, ‖u t x‖ ^ 2 ∂volume) ≤ ∫ x, ‖u 0 x‖ ^ 2 ∂volume) ∧
(∀ t ≤ (0 : ℝ), u t = u 0)
```

A time-dependent velocity field u belongs to `IsPhysicalNSSolution` if and only if for all $t \geq 0$:

1. $s \mapsto u(s, x)$ is pointwise differentiable at t for every $x \in \mathbb{R}^3$ (field of `IsPhysicalNSSolution`);
2. $x \mapsto u(t, x)$ is continuous and has finite kinetic energy $\int_{\mathbb{R}^3} \|u(t,x)\|^2 \, dx < \infty$ (field of `IsPhysicalNSSolution`);
3. u obeys the classical Leray L^2 energy inequality $\int_{\mathbb{R}^3} \|u(t,x)\|^2 \, dx \leq \int_{\mathbb{R}^3} \|u(0,x)\|^2 \, dx$ (field of `IsPhysicalNSSolution`);
4. $u(t) = u(0)$ for $t \leq 0$ (initial-value convention on $(-\infty, 0]$; field of `IsPhysicalNSSolution`).

The classical PDE inputs enter as explicit, transparent hypotheses:

- Kato local existence on physical solutions (`KatoLocalWitness`, carrying `phys:IsPhysicalNSSolutionu`) [7, 11];
- Beale–Kato–Majda continuation and parabolic smoothing (`bkm_regularity_pipeline`) [3, 4];
- Energy-Grönwall uniqueness of smooth physical solutions (`uniqueness_of_existence_times_of_energy_gronwall`) [11].

3 Geometric Layer: The Frohmanian Symplectic Tether

3.1 The Classical Arnold Lie–Poisson Bracket

On the coadjoint orbit $\mathcal{O}$ of the group $\mathrm{SDiff}(\mathbb{T}^3)$ of volume-preserving diffeomorphisms, the classical Kirillov–Kostant–Souriau / Arnold Lie–Poisson bracket [1, 2] (Lean: `ClassicalBracket`) is

$$\{F, G\}_{\mathrm{cl}}(\omega) = \int_{\mathbb{T}^3} \omega \cdot \left(\frac{\delta F}{\delta \omega} \times \frac{\delta G}{\delta \omega} \right) dV.$$

With kinetic-energy Hamiltonian $H(\omega) = \frac{1}{2} \int |u|^2 \, dV$ (where $\delta H / \delta \omega = u = \mathrm{BS}(\omega)$ is the Biot–Savart velocity reconstruction, formalized in `ArnoldGeometric.lean`), the Hamiltonian vector field of $\{F, H\}_{\mathrm{cl}}$ reproduces the ideal 3D Euler vorticity equation ($\nu = 0$ in (2); Lean: `classical_bracket_reproduces_Euler`).

3.2 Route A: The Projected Tether Kernel and Degeneracy

Definition 3.1 (Frohmanian tether kernel; Lean `TetherKernel`, `Pi_u`)**.** For a velocity field $u = \mathrm{BS}(\omega)$ with $\|u\|_{L^2} \neq 0$, let $\Pi_u = P_u^{\perp}$ be the L^2-orthogonal Gram–Schmidt projection onto the complement of $\mathrm{span}\{u\}$:

$$\Pi_u v = v - \frac{\langle v, u \rangle_{L^2}}{\|u\|_{L^2}^2} \, u$$

(Lean: `ForMathlib.Pi_u`, with $\Pi_u v = v$ when $\|u\|_{L^2} = 0$). With universal 3D Calderón–Zygmund constant $\kappa = C_{\mathrm{CZ}}(3) = \frac{3}{8\pi} \cdot 4\pi = \frac{3}{2}$ (Lean: `kappa_eq_three_halves` in `SymplecticTether.lean`), define the Frohmanian tether kernel

$$T(F, G)(\omega) = B(F, G)(\omega) = -\kappa \, \|\omega\|_{L^2}^2 \int_{\mathbb{T}^3} \left\langle \Pi_u \left(\frac{\delta F}{\delta \omega} \right)(x), \, \Pi_u \left(\frac{\delta G}{\delta \omega} \right)(x) \right\rangle dV,$$

where $\|\omega\|_{L^2}^2 = \int_{\mathbb{T}^3} |\omega|^2 \, dV$ is the enstrophy (Lean: `Enstrophy`), a constant on each coadjoint orbit. The full tethered bracket is $\{F, G\}_{\mathfrak{T}_F} = \{F, G\}_{\mathrm{cl}} + T(F, G)$.

Proposition 3.1 (Strict Degeneracy on Kinetic Energy; proved, `tetherKernel_degenerates_on_kinetic_energy`)**.** *Whenever $\delta H/\delta\omega = u$, $\nabla \cdot u = 0$, and $u \in L^2$, one has $\Pi_u u = 0$ (`projection_orthogonal_to_u`) and hence*

$$B(F, H)(\omega) = 0 \qquad \textit{for every functional } F.$$

Proof. If $\|u\|_{L^2} \neq 0$, $\Pi_u u = u - (\langle u, u\rangle_{L^2}/\|u\|_{L^2}^2)u = 0$, so the projected factor $\Pi_u(\delta H/\delta\omega)$ vanishes pointwise (`projection_orthogonal_to_u`). If $\|u\|_{L^2} = 0$, $u = 0$ a.e. and $\Pi_u 0 = 0$ (`Pi_u_zero`). Thus $\{F, H\}_{\mathfrak{T}_F} = \{F, H\}_{\mathrm{cl}}$ for all F: the reversible Euler vector field generated by H is preserved unmodified (`tetherKernel_degenerates_on_kinetic_energy`). □

3.3 Metriplectic Structure of the Tethered Bracket

The tether T of Definition 3.1 is a *symmetric* bilinear form in (F, G) (`tetherKernel_comm`, `tetherKernel_symmetric`) and is negative-semidefinite, $T(F, F) = -\kappa\|\omega\|_{L^2}^2 \int |\Pi_u \delta F/\delta\omega|^2 \leq 0$ (`ProducesControllableNegativeFeedback`). Hence

$$\{F, G\}_{\mathfrak{T}_F} = \underbrace{\{F, G\}_{\mathrm{cl}}}_{\text{antisymmetric, Lie–Poisson}} + \underbrace{T(F, G)}_{\text{symmetric, } \leq 0} \qquad (\texttt{tetherKernel_symmetric}, \texttt{tetherKernel_diagonal_no}$$

is a bracket of *metriplectic type* in the sense of Morrison [9] (symmetry and sign: `tetherKernel_symmetric`, `tetherKernel_diagonal_nonpos`; antisymmetry of the classical part: `classicalBracket_antisymm`). The full bracket is **not** antisymmetric, and **no Jacobi identity is claimed for it**; only its antisymmetric part $\{\cdot, \cdot\}_{\mathrm{cl}}$ is antisymmetric (`classicalBracket_antisymm`). Its Jacobi identity is cited from [1], not formalized. The symmetric part is nonzero off H (`tetherKernel_strict_neg_of_orthogonal_component`).

Energy conservation via (C2). Since $\{H, H\}_{\mathrm{cl}} = 0$ by antisymmetry (`classicalBracket_H_H`) and $T(H, H) = 0$ by kinetic-energy degeneracy (Proposition 3.1, condition (C2), `tetherKernel_degenerates_on_kinetic_energy`), the tethered flow $\dot{F} = \{F, H\}_{\mathfrak{T}_F}$ conserves H: $\dot{H} = \{H, H\}_{\mathrm{cl}} + T(H, H) = 0$. Here $T(H, H) = 0$ is `tetherKernel_degenerates_on_kinetic_energy`; $\{H, H\}_{\mathrm{cl}} = 0$ is `classicalBracket_H_H`.

3.4 Canonicity: Symmetric Uniqueness

In `Uniqueness.lean`, the canonicity theorem `uniqueness_of_minimal_tether` is stated for *symmetric* kernels $B'(\omega, F, G) = B'(\omega, G, F)$ and delegates to `uniqueness_of_minimal_symmetric_tether`: among symmetric bilinear kernels carrying the coadjoint-invariance (C1), kinetic-energy degeneracy (C2) and negative-feedback (C3) hypotheses, together with quadratic saturation (`SaturatesTetherQuadratic`) and polarization of B' and of `TetherKernel` (`Polarizes`), B' equals `TetherKernel`. The proof goes through polarization and saturation; the hypotheses (C1)–(C3) appear in the statement but are not used by the proof. The five-step route (`uniqueness_of_minimal_tether_core`) uses the density hypothesis, the enstrophy weight $\alpha(\omega, x) = -c\,\|\omega\|_{L^2}^2$, saturation and a non-degeneracy witness. `TetherKernel` itself is symmetric (`tetherKernel_symmetric`).

3.5 Coadjoint Invariance and the Conditional Jacobiator Lemma

In `SymplecticTether.lean`:

- **Abstract & push-forward coadjoint invariance:** `tether_coadjoint_invariance` and `tetherKernel_C1` verify `InvariantUnderCoadjointActionTetherKernel` *conditionally* on `SDiffActionHyp` and `TetherIntegralPreserved`. Here SDiff is an *abstract* group acting on the orbit type (a Lean `MulAction`); it is not constructed as the diffeomorphism group of $\mathbb{T}^3$, and volume preservation is a *hypothesis* on the field `diffeo` of `VolumePreservingCoadjointAction`, not a theorem. A concrete instance is given in `ConcreteSDiff.lean`: `ConcreteSDiff.SDiffConcrete` is a subgroup of `Equiv.Perm T3` whose elements g are measurable with measurable inverse, `MeasurePreserving` for `volume` in both directions, and preserve divergence-freeness under composition (this last property is part of the definition, not a theorem). It (`ConcreteSDiff.SDiffConcrete`) acts by scalar composition $g \cdot \omega = \omega \circ g^{-1}$, which is not the geometric push-forward $(Dg\,\omega) \circ g^{-1}$; no smoothness is imposed, and `T3` is `EuclideanSpace ℝ (Fin 3)`, not the quotient torus. For this action the field `casimir_invariant` of `SDiffActionHyp` is proved (`ConcreteSDiff.casimir_invariant`, via `MeasurePreserving.integral_comp'`), and `transitive_on_casimir_levels` remains an open hypothesis (`ConcreteSDiff.sdiffActionHyp_of_transitive`). To represent the full infinite-dimensional diffeomorphism push-forward $\omega \mapsto g_*\omega$ on $\mathrm{SDiff}(\mathbb{T}^3)$ without hiding the geometric bridge, we also define `ConjecturedSDiffPushforwardInvariance` and prove `tether_coadjoint_invariance_of_integral_eq`, showing that volume-preserving push-forward equivariance of the projected integrand yields coadjoint invariance of `TetherKernel` (the enstrophy weight is itself coadjoint-invariant).

- **Conditional Jacobiator lemma (`tethered_jacobi_identity`):** this lemma assumes, as explicit hypotheses, that the cyclic Jacobiator of `TetheredBracket` equals a cyclic Lie-pairing integral and that this integral vanishes, and concludes that the Jacobiator vanishes. Both hypotheses are open; the lemma does *not* establish a Jacobi identity for the full bracket, and none is claimed (a metriplectic bracket with nonzero symmetric part is not expected to satisfy one).

4 Local-Maximum Lemmas, the Cubic Comparison Ceiling, and Non-Vacuity

4.1 Local-Maximum Stretching and Transport Lemmas

Lemma 4.1 (Stretching at a spatial maximum; proved, `stretching_inner_le_at_max`)**.** *Let x satisfy $\|\omega(x)\| = \|\omega\|_{L^\infty}$ and suppose $\|\nabla u(x)\| \le C_{\mathrm{CZ}}\|\omega\|_{L^\infty}$ (`hCZ`). Then*

$$\langle \omega(x), (\omega \cdot \nabla)u(x)\rangle \le C_{\mathrm{CZ}}\,\|\omega\|_{L^\infty}^3.$$

Lemma 4.2 (Transport and viscosity at a spatial maximum; proved, `time_pairing_le_stretching_at_max`)**.** *At a spatial maximum where the convective transport pairing $\langle \omega, (u\cdot\nabla)\omega\rangle(x)$ vanishes and $\langle \omega, \Delta\omega\rangle(x) \le 0$, along any solution of (2) with $\nu \ge 0$,*

$$\langle \omega, \partial_t\omega\rangle(x) \le \langle \omega, (\omega\cdot\nabla)u\rangle(x).$$

4.2 The Quartic Lyapunov Vehicle and Cubic Comparison ODE

Caveat. The comparison results below are proved implications *from* a cubic differential inequality. Whether $M(t)$ satisfies such an inequality is open; it is the unresolved premise of Section 5.2; nothing in this section establishes it. Let φ solve the linear transport equation (`IsEnstrophyAccumulation`)

$$\partial_t\varphi + u\cdot\nabla\varphi = |\omega|^2, \tag{4}$$

and for mollified vorticity ω_ε define the tethered quartic Lyapunov functional (`LyapunovS`)

$$S_\varepsilon(t) = \int_{\mathbb{T}^3} \Big(\tfrac{1}{2}|\omega_\varepsilon|^2 + \tfrac{\kappa}{4}|\omega_\varepsilon|^4 \, \varphi_\varepsilon \Big) \, dV. \tag{5}$$

In `TetheredLyapunov.lean` and `CubicSourceCertificate.lean`, the Fréchet/time derivatives of the quadratic and quartic densities (`hasDerivAt_lyapunov_density`, `quartic_weight_source_eq`, `lyapunovSminus_source_density`) and the L^6 Hölder–Sobolev–Young absorption pipeline (`AnalyticPipeline.holder_I4_le_sobolev`, `stretching_bound_of_holder`) are proved in full.

Lemma 4.3 (Cubic comparison stability; proved, `comparison_ode_stability`)**.** *Let $C > 0$, $\kappa'' > 0$, and let $y : [0,\infty) \to \mathbb{R}$ be differentiable on $[0,\infty)$ with*

$$y'(s) \le C\, y(s)^2 - \kappa''\, y(s)^3 \qquad (s \ge 0).$$

Then for every $t \ge 0$,

$$y(t) \le \max\Big(y(0),\ \frac{C}{\kappa''} \Big).$$

Lemma 4.4 (Explicit majorant global bound; proved, `comparison_majorant_global_bound`)**.** *For $C, \kappa'' > 0$ and $y_0 \ge 0$, any solution $Y_{C,\kappa''}(t; y_0)$ of the comparison ODE $y' = Cy^2 - \kappa'' y^3$, $y(0) = y_0$ (`ComparisonODE`) satisfies $0 \le Y_{C,\kappa''}(t; y_0) \le \max(y_0, C/\kappa'')$ for all $t \ge 0$.*

Proposition 4.1 (Cubic rate $\Rightarrow$ uniform vorticity ceiling; proved implication, `riccati_ceiling_of_vorticity_di`; premise unresolved)**.** *If $M(t) = \|\omega(t)\|_{L^\infty}$ is differentiable on $[0,\infty)$ and satisfies $M'(t) \le C_{\mathrm{CZ}} M(t)^2 - \kappa'' M(t)^3$, then*

$$\sup_{t \ge 0} M(t) \le \max\Big(M(0), \frac{C_{\mathrm{CZ}}}{\kappa''} \Big).$$

4.3 Resolution of the Unpatched Time-Derivative Vacuity and Proof of Joint Satisfiability

In an earlier unpatched version of the Lean library, `NS_PDEup` was used in the uniqueness binder `huniq` without requiring time-differentiability of the comparison field v. Because Mathlib defines `deriv` to return 0 at non-differentiable points, the spatially constant step field $v(t) = \mathbf{1}_{\{t>1\}} e_0$ with $p = 0$ had formal time derivative 0 everywhere (including at the jump $t = 1$), thereby satisfying `NS_PDE` from $v(0) = 0$ while violating uniqueness at $t = 2$. We preserve this audit discovery as a formal theorem (`library_huniq_unsatisfiable_at_zero` in `ClayStatements.lean`) to document why the unpatched binder (`existenceTimes_unpatched`) needed repair.

The Completed Repair (`IsPhysicalNSSolution`). By threading `IsPhysicalNSSolution` through `KatoLocalWitness`, `existenceTimes`, `restart_of_kato_quantitative`, `FrohmanianTetherBound`, `FrohmanianTetherRate`, `conditional_regularity_of_open_hyps`, and `frohmanian_tether_theorem`, every solution and comparison field is required to be pointwise time-differentiable on $[0,\infty)$ and to obey the Leray L^2 energy inequality $\int \|v(t)\|^2 \le \int \|v(0)\|^2$. In `ClayStatements.lean`, we prove the complete non-vacuity and joint-satisfiability theorem suite:

1. `stepField_not_physical` : $\neg$ `IsPhysicalNSSolutionstepField` (the step field is not time-differentiable at $t = 1$, so it is excluded from the physical solution class);
2. `repaired_hInt`: the energy-difference integrability hypothesis of `ClosureBlueprint.conditional_regularity_of_open_hyps` holds unconditionally for all (u_0, ν);

3. `repaired_huniq_shift_at_zero` and `repaired_huniq_at_zero`: the repaired uniqueness hypotheses hold unconditionally at $u_0 = 0$ for every ν and every Kato witness w;

4. `frohmanianTetherBound_zero` and `frohmanianTetherRate_zero`: both `FrohmanianTetherBound0` and `FrohmanianTetherRate0` hold unconditionally at $u_0 = 0$ (since the Leray energy inequality forces any continuous physical solution with $u(0) = 0$ to vanish identically, giving $\omega \equiv 0$ and $M(t) \equiv 0$);

5. `conditional_regularity_nonvacuous_at_zero`: instantiates `ConditionalRegularity.conditional_regularity_of_open_hyps` at $u_0 = 0$ for every $\nu > 0$ **unconditionally**, proving in the Lean kernel that the hypotheses of the main theorem are jointly satisfiable and non-vacuous.

```
-- NS_Millennium_Proof/Modules/ClayStatements.lean
public theorem stepField_not_physical : ¬ IsPhysicalNSSolution stepField := ...
public theorem repaired_huniq_at_zero (ν : ℝ) (w : KatoLocalWitness 0 ν) : ... := ...
public theorem frohmanianTetherBound_zero (ν : ℝ) : FrohmanianTetherBound 0 ν := ...
public theorem frohmanianTetherRate_zero (ν : ℝ) : FrohmanianTetherRate 0 ν := ...

public theorem conditional_regularity_of_open_hyps_nonvacuous_at_zero (ν : ℝ) (hν : 0 < ν) :
    ∃ (u : TimeDependentVelocity),
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.IsSmooth (u t)) ∧
      (u 0 = 0) ∧
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.satisfies_NavierStokes u ν) ∧
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
      (∀ v : TimeDependentVelocity,
        (∀ t ≥ (0 : ℝ), ConditionalRegularity.IsSmooth (v t)) →
        v 0 = 0 →
        (∀ t ≥ (0 : ℝ), ConditionalRegularity.satisfies_NavierStokes v ν) →
        ∀ t ≥ (0 : ℝ), v t = u t) := ...
```

5 Original Conjectured Framework, Physical Motivation, and Bridges (Full Preservation from v8)

To ensure that every original idea, physical motivation, mathematical construction, and conjectured bridge from Benjamin Stanley Frohman's foundational manuscript (*Frohmanian_Tether_Final_Submission_v8.tex*) is preserved in full view alongside the kernel-verified Lean theorems, we present here the complete original conjectured programme and its exact correspondence to the Lean 4 formalization.

5.1 Conjectured Symplectic 2-Form and Modified Kirillov–Kostant–Souriau Bracket

In the original geometric programme, the coadjoint orbit $\mathcal{O} \subset (\mathfrak{sdiff}(\mathbb{T}^3))^*$ is endowed with the augmented bracket

$$\{F, G\}_{\mathfrak{T}_F} = \{F, G\}_{\mathrm{cl}} + B(F, G), \tag{6}$$

where the nonlocal Biot–Savart form of the tether correction is written

$$B_{\mathrm{BS}}(F, G) = -\kappa \iint_{\mathbb{T}^3 \times \mathbb{T}^3} |\omega(x)|^2 \, K_{\mathrm{BS}}(x - y) \left[\Pi_u\Big(\frac{\delta F}{\delta \omega}\Big)(x)\right] \cdot \left[\Pi_u\Big(\frac{\delta G}{\delta \omega}\Big)(y)\right] dy \, dx, \tag{7}$$

and its local projected metric realization (formalized in `FrohmanianTether.TetherKernel`, with the global enstrophy weight $\|\omega\|_{L^2}^2$ replacing the pointwise weight $|\omega(x)|^2$ of *v8*) is

$$B(F, G) = -\kappa \, \|\omega\|_{L^2}^2 \int_{\mathbb{T}^3} \left[\Pi_u\Big(\frac{\delta F}{\delta \omega}\Big)(x)\right] \cdot \left[\Pi_u\Big(\frac{\delta G}{\delta \omega}\Big)(x)\right] dx, \tag{8}$$

with $\kappa = C_{\mathrm{CZ}}(3) = 3/2$. On tangent variations $(\delta u, \delta v) \in T_\omega \mathcal{O}$, the associated tethered pairing (original programme; not formalized as stated, the formalized kernel is `TetherKernel`) is

$$\mathfrak{T}_F(\delta u, \delta v) = \int_{\mathbb{T}^3} \omega \cdot (\delta u \times \delta v)\, dV - \kappa \int_{\mathbb{T}^3} |\omega|^2 (\delta u \cdot \delta v)\, dV. \tag{9}$$

Because $\Pi_u u = 0$ for $u = \delta H/\delta\omega$, the tether is strictly degenerate along the kinetic-energy Hamiltonian ($B(F, H) \equiv 0$, proved in Proposition 3.1, `tetherKernel_degenerates_on_kinetic_energy`), so the reversible Hamiltonian vector field generated by H remains the exact classical Euler vector field. As explained in Section 3.3, the augmented bracket (6) is metriplectic-type, not a Poisson bracket, and the 2-form language of *v8* is retained only historically.

5.2 The Damping Coefficient Vanishes: Explicit Caveat

The original programme derived a damping term for $M(t)$ from the tether; that derivation is not established, as shown below. That derivation does not survive. Let $\mathcal{M}$ be any functional (`FunctionalDerivative` gives its derivative) (e.g. a smoothed version of $M(t) = \|\omega\|_{L^\infty}$). Along the tethered flow,

$$\dot{\mathcal{M}} = \{\mathcal{M}, H\}_{\mathrm{cl}} + T(\mathcal{M}, H), \qquad (\texttt{tetherKernel_left_H_zero})$$

and $T(\mathcal{M}, H) = T(H, \mathcal{M}) = 0$ by symmetry of T and (C2) (`tetherKernel_left_H_zero`), because $\Pi_u(\delta H/\delta\omega) = \Pi_u u = 0$. Hence the damping coefficient contributed by the tether is exactly **0** (`tetherKernel_left_H_zero`): the tethered evolution of every functional coincides with its Euler evolution, and the negative diagonal values $T(F, F) \le 0$ never enter the dynamics generated by H.

Projection lower-bound gap. Any attempt to extract damping from the diagonal $T(F, F)$ (as in `tethered_stretching_of_weight`) requires a lower bound $M^4 \le \|\omega\|_{L^2}^2 \int |\Pi_u \delta F/\delta\omega|^2$ (`hweight`) for a test functional tied to M. No such lower bound on the projected component is known, and Π_u annihilates exactly the direction $u = \delta H/\delta\omega$ that drives the evolution; this gap is open.

Open status. Consequently the cubic rate (R) (`FrohmanianTetherRate`) and the closing bound (`FrohmanianTetherBound`, `PeriodicTetherClosingBound`) are **open assumptions**. Candidate repair routes—for example an explicit dissipative (metriplectic) bracket $[F, H]_{\mathrm{diss}}$ with a nonzero $[M, H]$, a time-integrated or non-local mechanism, or a scale-compensated coefficient (Section 6.1)—are listed only as candidates and remain open; none is carried out here.

5.3 Conjectured Unmodified-Tether Pairing Bridge at Spatial Maxima

Caveat. By Section 5.2 the tether contributes no damping; the results in this subsection are proved implications from explicit, open hypotheses (`ConjecturedUnmodifiedTetherPairing`, `hweight`) and do not supply (R). The central analytic bridge of the original programme connects the quadratic feedback of the tether kernel to the pointwise vortex-stretching pairing (`reduced_pairing_of_unmodified_and_tether`, `tethered_stretching_of_weight`) at a spatial maximum x_* of $|\omega|$ (where $\|\omega(x_*)\| = M(t) = \|\omega\|_{L^\infty}$). In `SymplecticTether.lean`, this bridge is formalized without `sorry` via two complementary theorems:

1. **Weighted tether-stretching absorption (`tethered_stretching_of_weight`):** Under the weight dominance condition (`hweight`)

$$M(t)^4 \le \|\omega_{\mathrm{orb}}\|_{L^2}^2 \int_{\mathbb{T}^3} \left|\Pi_u(\delta F/\delta\omega)(y)\right|^2 dy,$$

the sum of the pointwise stretching pairing $\langle \omega(x_*), (\omega \cdot \nabla)u(x_*)\rangle$ and the diagonal tether value $B(F,F)(\omega_{\mathrm{orb}})$ is bounded by

$$\langle \omega(x_*), (\omega \cdot \nabla)u(x_*)\rangle + B(F,F)(\omega_{\mathrm{orb}}) \le (CM(t) - \kappa\, M(t)^2)\, M(t)^2,$$

where C is the strain constant in `hCZ`. This implication is proved in `SymplecticTether.lean` with zero `sorry`s.

2. **Unmodified stretching reduction (`reduced_pairing_of_unmodified_and_tether`):** The bridge proposition `ConjecturedUnmodifiedTetherPairing` (stated along a Navier–Stokes solution at time t and a spatial maximum x) asserts the bound $\langle \omega(x), (\omega\cdot\nabla)u(x)\rangle \le (CM - \kappa M^2)M^2$ under the local-maximum hypotheses. The theorem `reduced_pairing_of_unmodified_and_tether` derives this bound from the explicit cancellation hypothesis `hcancels` ($\langle \omega, (\omega \cdot \nabla)u\rangle + \kappa M^4 \le CM^3$), which is itself the open content.

```
-- NS_Millennium_Proof/Modules/SymplecticTether.lean (sorry-free)
@[expose] public def ConjecturedUnmodifiedTetherPairing
    (u : TimeDependentVelocity) (p : TimeDependentPressure) (ν : ℝ)
    (t : ℝ) (x : T3) (C M : ℝ) : Prop :=
  0 ≤ t →
  NS_PDE u p ν →
  VorticityTransportRegularity u p t x →
  DifferentiableAt ℝ (u t) x →
  PointwiseStrainSupBound (u t) (vorticity (u t)) C x →
  inner ℝ (vorticity (u t) x) (convective (u t) (vorticity (u t)) x) = 0 →
  inner ℝ (vorticity (u t) x) (laplacian (vorticity (u t)) x) ≤ 0 →
  0 ≤ ν →
  (‖vorticity (u t) x‖ = M ∧ M = vorticity_sup_norm (vorticity (u t))) →
  ProducesControllableNegativeFeedback TetherKernel →
  kappa = CalderonZygmundConstant3D →
  inner ℝ (vorticity (u t) x) (convective (vorticity (u t)) (u t) x) ≤
    (C * M - kappa * M ^ 2) * M ^ 2

public theorem reduced_pairing_of_unmodified_and_tether
    (u : TimeDependentVelocity) (p : TimeDependentPressure) (ν : ℝ)
    (t : ℝ) (_ht : 0 ≤ t) (x : T3)
    (_hNS : NS_PDE u p ν)
    (_hreg : VorticityTransportRegularity u p t x)
    (_hu : DifferentiableAt ℝ (u t) x)
    (C M : ℝ)
    (_hCZ : ‖fderiv ℝ (u t) x‖ ≤ C * vorticity_sup_norm (vorticity (u t)))
    (_htransp : inner ℝ (vorticity (u t) x)
      (convective (u t) (vorticity (u t)) x) = 0)
    (_hvisc : inner ℝ (vorticity (u t) x)
      (laplacian (vorticity (u t)) x) ≤ 0)
    (_hν : 0 ≤ ν)
    (_hmax : ‖vorticity (u t) x‖ = M ∧
      M = vorticity_sup_norm (vorticity (u t)))
    (_htether : ProducesControllableNegativeFeedback TetherKernel)
    (_hκ : kappa = CalderonZygmundConstant3D)
    (hcancels : inner ℝ (vorticity (u t) x)
        (convective (vorticity (u t)) (u t) x) +
      kappa * M ^ 4 ≤ C * M ^ 3) :
    inner ℝ (vorticity (u t) x)
        (convective (vorticity (u t)) (u t) x) ≤
      (C * M - kappa * M ^ 2) * M ^ 2 :=
  (reduced_pairing_iff_cancels_kappa_M4
    (inner ℝ (vorticity (u t) x) (convective (vorticity (u t)) (u t) x))
    C kappa M).mpr hcancels
```

```
public theorem tethered_stretching_of_weight
    (ωField uField : VelocityField) (x : T3) (C M : ℝ)
    (hu : DifferentiableAt ℝ uField x)
    (hCZ : ‖fderiv ℝ uField x‖ ≤ C * vorticity_sup_norm ωField)
    (hmax : ‖ωField x‖ = M ∧ M = vorticity_sup_norm ωField)
    (orbit : CoadjointOrbit) (testF : Functional)
    (hweight : M ^ 4 ≤ Enstrophy orbit * ∫ y,
```

```
        ‖Pi_u (velocity_from_vorticity orbit)
            (FunctionalDerivative testF orbit) y‖ ^ 2 ∂volume) :
  inner ℝ (ωField x) (convective ωField uField x) +
      TetherKernel orbit testF testF ≤
    (C * M - kappa * M ^ 2) * M ^ 2 := ...
```

5.4 Quartic Lyapunov Absorption and Quadratic Riccati Heuristic vs. Cubic Ceiling

In Section 3 and Appendix A (Item 10) of *Frohmanian_Tether_Final_Submission_v8*, two related ODE reductions (historical, not formalized as stated) were introduced for the evolution of $M_\varepsilon(t) = \|\omega_\varepsilon(t)\|_{L^\infty}$:

1. **The Quartic Lyapunov L^6 Absorption and Cubic Comparison ODE (Formalized in Lean):** Differentiating the mollified quartic Lyapunov functional

$$S_\varepsilon(t) = \int_{\mathbb{T}^3} \Big(\frac{1}{2}|\omega_\varepsilon|^2 + \frac{\kappa}{4}|\omega_\varepsilon|^4 \varphi_\varepsilon \Big)\, d\lambda$$

along the vorticity transport equation and applying Hölder's inequality with conjugate exponents $(3/2, 3)$ (`AnalyticPipeline.holderConjugate_three_halves_three`, `holder_I4`), the 3D Sobolev embedding $H^1 \hookrightarrow L^6$ (`holder_I4_le_sobolev`), and Young's absorption parameter $\varepsilon = \kappa/4$ yields

$$\frac{dS_\varepsilon}{dt} \le C_{\mathrm{abs}}\big(1 + M_\varepsilon(t)^3 \|\varphi_\varepsilon\|_{L^\infty}^{3/2}\big) - \kappa' \int_{\mathbb{T}^3} |\omega_\varepsilon|^6\, d\lambda, \tag{10}$$

which motivates the autonomous cubic comparison ODE (`FrohmanianTetherRate`, `ComparisonODE`)

$$y'(t) = C\, y(t)^2 - \kappa''\, y(t)^3, \qquad y(0) = M(0). \tag{11}$$

As proved in Lemma 4.3 (`comparison_ode_stability`), the equilibrium $y_* = C/\kappa''$ is globally asymptotically stable on $(0, \infty)$, yielding the ceiling $M(t) \le \max(M(0), C/\kappa'')$ for every trajectory that satisfies the inequality. Whether $M(t)$ does satisfy it is open; it is the unresolved premise of Section 5.2.

2. **The Original Quadratic Riccati Heuristic (from Section 3 of v8; depends on the unresolved damping premise):** In Section 3 of *v8*, an alternative constant-source quadratic Riccati heuristic (not proved) was written:

$$\frac{d}{dt} M_\varepsilon(t) \le C - \kappa'\, M_\varepsilon(t)^2 \qquad \text{(heuristic, not proved)} \tag{12}$$

with comparison trajectory $y(t) = \sqrt{C/\kappa'} \tanh\big(\sqrt{C\kappa'}\, t + \phi_0\big)$ for $y(0) < \sqrt{C/\kappa'}$ (and coth for $y(0) > \sqrt{C/\kappa'}$), both bounded above by $\max\big(M(0), \sqrt{C/\kappa'}\big)$. In our Lean 4 formalization, the cubic differential inequality (11) from Appendix A (Item 10) of *v8* is adopted as the canonical formulation (`FrohmanianTetherRate`) because its phase-plane barrier proof (`comparison_ode_stability`) handles both small and large initial data uniformly without switching between tanh and coth branches.

5.5 Metriplectic Completion, Onsager's Conjecture, and Holographic Fluid–Gravity Motivation

The broader physical motivation of the Frohmanian programme in Sections 4–6 and Appendix A (Items 11–12) of *v8* encompasses three structural themes:

- **Metriplectic Viscous Completion [9]:** The tethered bracket is already metriplectic-type, but its symmetric part is degenerate on H and so dissipates nothing (Section 5.2; `tetherKernel_left_H_zero`). Viscous dissipation $\nu\Delta u$ would have to be represented by a separate symmetric dissipative bracket $[F, H]_{\text{diss}}$ (an open candidate, not constructed here; this remains open):
$$\frac{dF}{dt} = \{F, H\}_{\mathfrak{T}_F} + [F, H]_{\text{diss}}.$$
In the Lean 4 statement of Theorem 1.1, *if* a vorticity ceiling $M(t) \leq Y$ were available, classical Beale–Kato–Majda continuation and parabolic regularity would upgrade the solution to C^∞; such a ceiling is not established here (BKM and parabolic regularity are cited, not formalized).

- **Relation to Onsager's Conjecture [6]:** Onsager's conjecture (proved by Isett [6]) states that weak Euler solutions conserve kinetic energy for Hölder regularity $\alpha > 1/3$, whereas anomalous dissipation can occur for $\alpha \leq 1/3$. *If* a vorticity ceiling held (an open premise, Section 5.2), smooth solutions would remain in the Lipschitz/smooth class ($\alpha = 1$); this is not claimed.

- **Holographic Fluid–Gravity Dual and Cosmic Censorship:** Under the AdS/CFT fluid–gravity correspondence, a finite-time vorticity blow-up on the 3D boundary manifold corresponds to the formation of a naked curvature singularity in the 4D asymptotically AdS bulk spacetime. Penrose's cosmic censorship conjecture in the bulk therefore provides a geometric dual heuristic motivating the existence of an ultraviolet vorticity ceiling on the boundary.

5.6 Complete Preservation of the 14-Point Programme from Appendix A of *v8*

For historical completeness and traceability, Table 1 maps each of the 14 mandatory items from Appendix A of *Frohmanian_Tether_Final_Submission_v8.tex* to its exact realization in the Lean 4 formalization `NS_Millennium_Proof`.

6 The Periodic Clay (B) Statement and Scaling Analysis

Fefferman's problem (B) [5] asks (background, not formalized here as such), on $\mathbb{T}^3$ with $f \equiv 0$, for smooth spatially periodic solutions (u, p) on $\mathbb{R}^3 \times [0, \infty)$. It is formalized in `ClayStatements.lean`:

```
@[expose] public def Clay_B_exact : Prop :=
  ∀ (ν : ℝ) (u₀ : VelocityField),
    0 < ν →
    ContDiff ℝ (⊤ : ℕ∞) u₀ →
    (∀ x, div u₀ x = 0) →
    IsZ3Periodic u₀ →
    ∃ (u : TimeDependentVelocity) (p : TimeDependentPressure),
      IsPeriodicSmoothSolution ν u₀ (fun _ _ => 0) u p
```

```
@[expose] public def IsPeriodicSmoothSolution (ν : ℝ) (u₀ : VelocityField)
    (f : ℝ → T3 → EuclideanSpace ℝ (Fin 3))
    (u : TimeDependentVelocity) (p : TimeDependentPressure) : Prop :=
  ContDiffOn ℝ ∞ (fun q : ℝ × T3 => u q.1 q.2) (Set.Ici (0 : ℝ) ×ˢ Set.univ) ∧
  ContDiffOn ℝ ∞ (fun q : ℝ × T3 => p q.1 q.2) (Set.Ici (0 : ℝ) ×ˢ Set.univ) ∧
  navier_stokes_eq_Ici u p ν f ∧
  u 0 = u₀ ∧
  ∀ t ≥ (0 : ℝ), IsZ3Periodic (u t) ∧ IsZ3Periodic (p t)
```

#	Item in *v8* Appendix A	Lean 4 Module / Declaration	Formal Status
1	ZFC Foundations, $\mathbb{R}^3$, $\mathbb{T}^3$, L^2, H^s Sobolev spaces	Mathlib foundations + `NS_Equations.lean`	Kernel-verified in Lean 4 / Mathlib
2	Unmodified 3D Navier–Stokes & vorticity transport (2)	`NS_PDE`, `vorticity_transport_equation`, `IsPhysicalNSSolution`	Kernel-verified (0 `sorry`)
3	Phase space: coadjoint orbit $\mathcal{O}$ of SDiff	`VorticityField`, `Functional`, `CoadjointAction`	Defined in `ArnoldGeometric.lean`, `SymplecticTether.lean`
4	Classical Arnold Lie–Poisson bracket $\{F, G\}_{\mathrm{cl}}$	`ClassicalBracket`, `classical_bracket_reproduces_Euler`	Kernel-verified (0 `sorry`)
5	Explicit Frohmanian Tether $B(F, G)$, Π_u, $\kappa = 3/2$, degeneracy	`TetherKernel`, `Pi_u`, `kappa_eq_three_halves`, `tetherKernel_degenerates_on_kinetic_energy`	Kernel-verified (0 `sorry`)
6	Coadjoint invariance of quadratic correction	`tether_coadjoint_invariance`, `tether_coadjoint_invariance_of_integral_eq`	Kernel-verified (abstract & conditional push-forward)
7	Reduced Jacobi identity (9-term expansion + cocycle)	`tethered_jacobi_identity`	Conditional lemma only; no Jacobi identity claimed for the metriplectic-type bracket
8	Canonicity & Uniqueness Theorem (5-step)	`uniqueness_of_minimal_tether`, `uniqueness_of_minimal_symmetric_tether`, `uniqueness_of_minimal_tether_core`	Kernel-verified (symmetric form)
9	Variational principle & Leray projector orthogonality	`projector_orthogonality`, `pressure_energy_pairing_vanishes`	Kernel-verified (0 `sorry`)
10	Quartic Lyapunov S_ε, scalar φ, L^6 absorption, cubic ODE $y' = Cy^2 - \kappa'' y^3$, BKM	`LyapunovS`, `comparison_ode_stability`, `frohmanian_tether_theorem`, `conditional_regularity_of_open_hyps`	Kernel-verified conditional implication + non-vacuity at $u_0 = 0$; premise (R) open, damping coefficient $= 0$
11	Relation to Onsager's conjecture ($\alpha = 1$ persistence)	Section 5 of this paper	Preserved in manuscript as physical motivation
12	Metriplectic & holographic fluid–gravity extensions	`Skeleton/PaperOverview.lean` (outside the default build target) & Section 5	Preserved in manuscript & overview module
13	Technical Appendix: invariance, MWR, mollifiers, pressure	`NS_Equations.lean`, `TetheredLyapunov.lean`, `AnalyticPipeline.lean`	Kernel-verified lemmas (537 theorems in the default target)
14	Conclusion: non-circular conditional architecture	`ConditionalRegularity.lean`, `ClayStatements.lean`	Checked by `scripts/AxiomAudit.lean`

Table 1: Complete mapping of the 14-point programme of *Frohmanian_Tether_Final_Submission_v8* to the Lean 4 formalization.

The reduction from `Clay_B_exact` to the periodic closing bound `PeriodicTetherClosingBound` and classical periodic BKM continuation `PeriodicBKMContinuation` is proved axiom-clean:

```
public theorem clay_B_exact_of_periodic_tether_bound
    (hclose : ∀ ν > (0 : ℝ), PeriodicTetherClosingBound ν)
    (hBKM : ∀ ν > (0 : ℝ), PeriodicBKMContinuation ν) : Clay_B_exact :=
  fun ν u₀ hν hs hd hp => hBKM ν hν u₀ hd hs hp (hclose ν hν u₀ hd hs hp)
```

6.1 Scaling Analysis and Open Analytic Questions

While the conditional implication Theorem 1.1 is proved and non-vacuous in Lean 4, its premise (R) is unresolved: the tether's damping coefficient is 0 (Section 5.2), and discharging the cubic rate hypothesis (R) (`FrohmanianTetherRate`) unconditionally from the unmodified vorticity equation (2) remains an open mathematical problem for the following structural reasons:

1. **Parabolic scaling dimension of κ'':** Under the Navier–Stokes rescaling (3), $M'(t)$ scales as λ^4, $C_{\mathrm{CZ}}M(t)^2$ scales as λ^4, and $\kappa''M(t)^3$ scales as λ^6 (`navier_stokes_vorticity_dt_scaling_exponent`, `unnormalized_quartic_scaling_mismatch`). A fixed constant $\kappa'' > 0$ therefore (`unnormalized_quartic_scaling_mismatch`) carries the dimension of time and cannot be formed from viscosity ν alone ($[\nu] = L^2T^{-1}$) without incorporating a solution-dependent or scale-compensating functional (see `SCALE_INVARIANT_TETHER.md` and `CLAY_B_T3.md`).

2. **Pointwise strain at spatial maxima (`HWEIGHT_OBSTRUCTION.md`):** Because $B(F, H) = 0$ by Proposition 3.1 and B is symmetric, $B(M, H) = 0$ for every M: the tether adds no damping term to (2) or to any functional. Consequently, the pointwise inequality $\langle\omega, (\omega\cdot\nabla)u\rangle(x_*) \le C_{\mathrm{CZ}}M^3 - \kappa M^4$ (`ConjecturedUnmodifiedTetherPairing`) and the integral weight dominance condition $M^4 \le \|\omega\|_{L^2}^2 \int |\Pi_u \delta F|^2$ (`hweight`) do not hold for arbitrary divergence-free fields with vanishing stretching at the maximum when $M > C_{\mathrm{CZ}}/\kappa$. Closing (R) for the unmodified PDE therefore requires a non-local or time-integrated mechanism beyond pointwise evaluation at a single time slice.

3. **Calderón–Zygmund L^∞ logarithm:** Singular integral operators (such as $\nabla u = \nabla \mathrm{BS}(\omega)$) are not bounded from L^∞ to L^∞ without a logarithmic Sobolev correction $\|\nabla u\|_{L^\infty} \lesssim \|\omega\|_{L^\infty}(1 + \log^+(\|\omega\|_{H^s}/\|\omega\|_{L^\infty}))$ [3] (cited, not formalized in the Lean library). In the pointwise local-maximum lemma, $\|\nabla u(x)\| \le C_{\mathrm{CZ}}\|\omega\|_{L^\infty}$ enters as the explicit hypothesis `hCZ` of `stretching_inner_le_at_max`.

7 Scope Table: Complete Status of Every Component

Component	Lean Declaration / Location	Verification Status
G1 Tether kernel	`TetherKernel`, `Pi_u`	Defined & symmetric (`tetherKernel_symmetric`)
G2 Degeneracy $B(F, H) = 0$	`tetherKernel_degenerates_on_kinetic_energy`, `Combined.tetherKernel_C2_of_KEDeriv`	**Proved axiom-clean, conditional** on $\delta H/\delta\omega = u$ (`KineticEnergyDerivIsVelocity`, open), $\nabla\cdot u = 0$, finite energy
A1 Canonicity (symmetric)	`uniqueness_of_minimal_tether`	**Proved axiom-clean** (symmetric binder; delegates to the symmetric theorem)
A1 Symmetric uniqueness	`uniqueness_of_minimal_symmetric_tether`	**Proved axiom-clean**

Component	Lean Declaration / Location	Verification Status
A3/C1 Coadjoint invariance	`tether_coadjoint_invariance`, `Combined.tetherKernel_C1_of_casimir`, `Combined.tetherKernel_C1_of_volumePreserving`	**Proved axiom-clean, conditional** on `SDiffActionHyp` (field `casimir_invariant`) and `TetherIntegralPreserved`; abstract SDiff
A5/C3 Negative feedback	`ProducesControllableNegativeFeedback`	**Proved axiom-clean**
Bracket structure	`TetheredBracket`	Metriplectic-type (antisymmetric + symmetric ≤ 0); **not** antisymmetric; no Jacobi claimed
Conditional Jacobiator lemma	`tethered_jacobi_identity`	Proved from two open hypotheses; not a Jacobi identity
Tether damping coefficient	$T(M, H) = 0$ (symmetry + C2)	**Zero**; cubic rate (R) and closing bound are **open assumptions**
G4 Stretching at spatial max	`stretching_inner_le_at_max`	**Proved axiom-clean** (given `hCZ`)
G4 Transport/viscosity at max	`time_pairing_le_stretching_at_max`	**Proved axiom-clean**
Conjectured pairing bridge	`ConjecturedUnmodifiedTetherPairing`, `reduced_pairing_of_unmodified_and_tether`	**Proved axiom-clean (0 `sorry`)** from explicit bridge hypothesis
Weighted stretching lemma	`tethered_stretching_of_weight`	**Proved axiom-clean (0 `sorry`)** from `hweight`
N1 Lyapunov vehicle S_ε, φ	`LyapunovS`, `IsEnstrophyAccumulation`	**Proved axiom-clean**
N2 Cubic ODE comparison	`comparison_ode_stability`	**Proved axiom-clean**
N2 Explicit majorant bound	`comparison_majorant_global_bound`	**Proved axiom-clean**
N3 Rate $\Rightarrow$ ceiling	`riccati_ceiling_of_vorticity_di`	**Proved axiom-clean** implication; premise unresolved
N4 Physical solution class	`IsPhysicalNSSolution`, `isPhysicalNSSolution_zero`	**Proved axiom-clean**
N4 Main conditional theorem	`conditional_regularity_of_open_hyps`	**Proved axiom-clean & verified non-vacuous** as an implication; depends on unresolved premise (R); no regularity claim
N4 Assembled tether theorem	`frohmanian_tether_theorem`	**Proved axiom-clean** as an implication; depends on unresolved premise (R)
Non-vacuity & joint satisfiability	`stepField_not_physical`, `repaired_hInt`, `repaired_huniq_at_zero`, `frohmanianTetherBound_zero`, `frohmanianTetherRate_zero`, `conditional_regularity_nonvacuous_at_zero`	**Proved unconditionally at** $u_0 = 0$ **(axiom-clean)**
Historical unpatched audit	`library_huniq_unsatisfiable_at_zero`, `closing_bound_uniqueness_family_unsatisfiable`	**Proved axiom-clean** (documents why `IsPhysicalNSSolution` was added)
N5 Clay (B) exact statement	`Clay_B_exact`, `IsPeriodicSmoothSolution`	Formalized in `ClayStatements.lean`

Component	Lean Declaration / Location	Verification Status
N5 Clay (B) conditional reduction	`clay_B_exact_of_periodic_tether_bound`	**Proved axiom-clean** (conditional on `PeriodicTetherClosingBound`)

8 Certificate

- Builds clean: `lake build NS_Millennium_Proof` succeeds.
- No `sorry`.
- Standard axioms only: `propext`, `Classical.choice`, `Quot.sound`.
- Geometric layer proved: C1, C2, C3, canonicity, Casimir conservation and $T(M,H)=0$ are proved in `Combined.combined_conditional_capstone` from its listed hypotheses.
- Analytic layer stated as open assumptions: `FrohmanianTetherRate`, `FrohmanianTetherClosingBound`, Kato local existence, BKM continuation and Kato uniqueness are hypotheses of `Combined.combined_conditional_capstone`, not theorems.

The hypotheses of `Combined.combined_conditional_capstone` are: `SDiffActionHyp`, `TetherIntegralPreserved`, `KineticEnergyDerivIsVelocity`, $\nabla \cdot u = 0$ and finite energy on the orbit, `IsCoadjointMotion`, symmetry of B, `h_proj`, `h_dens`, `h_weight`, `h_sat`, `h_nondeg`, smooth divergence-free finite-energy u_0, $\nu > 0$, `KatoLocalWitness`, `FrohmanianTetherRate`, `FrohmanianTetherClosingBound`, `PhysicalBKMContinuationCriterionFor` and `KatoUniquenessOnExistenceTimes`. Under the symmetric C2 design the damping coefficient is 0 (`tetherKernel_left_H_zero`), so nothing proved here supplies the rate or closing-bound hypotheses.

A Exact Lean Statements

Quoted verbatim from `NS_Millennium_Proof` (537 theorems, 0 `sorry`, 0 nonstandard axioms).

Physical solution class (`NS_Equations.lean`).

```
-- NS_Millennium_Proof/Modules/NS_Equations.lean
@[expose] public def IsPhysicalNSSolution (u : TimeDependentVelocity) : Prop :=
  (∀ t ≥ (0 : ℝ), ∀ x : T3, DifferentiableAt ℝ (fun s => u s x) t) ∧
  (∀ t ≥ (0 : ℝ), Continuous (u t)) ∧
  (∀ t ≥ (0 : ℝ), Integrable (fun x : T3 => ‖u t x‖ ^ 2)) ∧
  (∀ t ≥ (0 : ℝ), (∫ x, ‖u t x‖ ^ 2 ∂volume) ≤ ∫ x, ‖u 0 x‖ ^ 2 ∂volume) ∧
  (∀ t ≤ (0 : ℝ), u t = u 0)
```

Main conditional theorem (`ConditionalRegularity.lean`).

```
-- NS_Millennium_Proof/Modules/ConditionalRegularity.lean
public theorem conditional_regularity_of_open_hyps
    (u0 : TimeDependentVelocity)
    (h_div_free : ∀ t x, div (u0 t) x = 0)
    (h_smooth : IsSmooth (u0 0))
    (h_finite_energy : Integrable (fun x : T3 => ‖u0 0 x‖ ^ 2))
    (ν : ℝ) (h_ν_pos : ν > 0)
    (w : KatoLocalWitness (u0 0) ν)
    (hRiccati : FrohmanianTetherBound (u0 0) ν)
    (hcrit : TetheredLyapunov.PhysicalBKMContinuationCriterionFor (u0 0) ν)
    (huniq : TetheredLyapunov.KatoUniquenessOnExistenceTimes (u0 0) ν w) :
```

```
  ∃ (u : TimeDependentVelocity),
    (∀ t ≥ (0 : ℝ), IsSmooth (u t)) ∧
    (u 0 = u0 0) ∧
    (∀ t ≥ (0 : ℝ), satisfies_NavierStokes u ν) ∧
    (∀ t ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
    (∀ v : TimeDependentVelocity,
      (∀ t ≥ (0 : ℝ), IsSmooth (v t)) →
      v 0 = u0 0 →
      (∀ t ≥ (0 : ℝ), satisfies_NavierStokes v ν) →
      ∀ t ≥ (0 : ℝ), v t = u t) := by
  obtain ⟨hNS, hu0, hsm, hnn, _hdiv⟩ :=
    TetheredLyapunov.conditional_regularity_of_bkm_uniqueness (u0 0) ν h_ν_pos
      w (TetheredLyapunov.physicalFiniteTimeVorticityBound_of_uniform
          (TetheredLyapunov.physicalUniformVorticityCeiling_of_tetherBound hRiccati))
      hcrit huniq
  refine ⟨w.u, hsm, hu0, ?ns, hnn, ?uniq⟩
  · intro _t _ht
    exact ⟨w.phys, w.p, hNS⟩
  · intro v hsmv hv0 hNSv t ht
    obtain ⟨hphys_v, q, hq⟩ := hNSv 0 le_rfl
    have ht1 : 0 < t + 1 := by linarith
    have hmem : t + 1 ∈ existenceTimes (u0 0) ν :=
      mem_existenceTimes (u0 0) ν ht1 v q hv0 hq (fun s hs => hsmv s hs.1) hphys_v
    exact (huniq (t + 1) v q hmem hv0 hq (fun s hs => hsmv s hs.1) hphys_v
      t ⟨ht, le_of_lt (lt_add_one t)⟩).symm
```

Assembled Frohmanian tether theorem (`ConditionalRegularity.lean`).

```
-- NS_Millennium_Proof/Modules/ConditionalRegularity.lean
public theorem frohmanian_tether_theorem
    (hKato : ∀ u₀ ν, ContDiff ℝ (⊤ : ℕ∞) u₀ → (∀ x, div u₀ x = 0) →
      Integrable (fun x : T3 => ‖u₀ x‖ ^ 2) → 0 < ν →
      KatoLocalWitness u₀ ν)
    (hDI : ∀ u₀ ν, FrohmanianTetherRate u₀ ν)
    (τ : ℝ) (hτ : 0 < τ)
    (hTlb : ∀ u₁ ν (hsm : ContDiff ℝ (⊤ : ℕ∞) u₁)
      (hdiv : ∀ x, div u₁ x = 0)
      (hE : Integrable (fun x : T3 => ‖u₁ x‖ ^ 2))
      (hν : 0 < ν),
      τ ≤ (hKato u₁ ν hsm hdiv hE hν).T)
    (hInt : ∀ u₀ ν (_hν : 0 < ν) (T : ℝ) (v : TimeDependentVelocity)
      (q : TimeDependentPressure),
      T ∈ existenceTimes u₀ ν →
      v 0 = u₀ →
      NS_PDE v q ν →
      (∀ t ∈ Set.Icc (0 : ℝ) T, ContDiff ℝ (⊤ : ℕ∞) (v t)) →
      IsPhysicalNSSolution v →
      ∀ t ∈ Set.Icc (0 : ℝ) T,
        Integrable (fun x : T3 => ‖v t x‖ ^ 2))
    (huniq_shift : ∀ u₀ ν (_hν : 0 < ν) (T : ℝ) (v : TimeDependentVelocity)
      (q : TimeDependentPressure),
      T ∈ existenceTimes u₀ ν →
      v 0 = u₀ →
      NS_PDE v q ν →
      (∀ t ∈ Set.Icc (0 : ℝ) T, ContDiff ℝ (⊤ : ℕ∞) (v t)) →
      IsPhysicalNSSolution v →
      ∀ t ∈ Set.Icc (0 : ℝ) T,
        ∀ (w' : KatoLocalWitness (v t) ν),
          ∀ s ∈ Set.Icc (0 : ℝ) w'.T, v (t + s) = w'.u s)
    (huniq : ∀ u₀ ν (w : KatoLocalWitness u₀ ν),
      ∀ (T' : ℝ) (v : TimeDependentVelocity) (q : TimeDependentPressure),
        T' ∈ existenceTimes u₀ ν →
        v 0 = u₀ →
        NS_PDE v q ν →
        (∀ t ∈ Set.Icc (0 : ℝ) T', ContDiff ℝ (⊤ : ℕ∞) (v t)) →
        IsPhysicalNSSolution v →
        ∀ t ∈ Set.Icc (0 : ℝ) T', w.u t = v t) :
  ∃ (𝔗_F : CoadjointOrbit → Functional → Functional → ℝ),
    (∀ F ω,
      FunctionalDerivative KineticEnergyHamiltonian ω =
          velocity_from_vorticity ω →
```

```
    (∀ x, div (velocity_from_vorticity ω) x = 0) →
    Integrable (fun y => ‖velocity_from_vorticity ω y‖ ^ 2) →
    TetheredBracket F KineticEnergyHamiltonian ω =
      ClassicalBracket F KineticEnergyHamiltonian ω) ∧
  (∀ (SDiff : Type) [Group SDiff] [MulAction SDiff CoadjointOrbit] (B : CoadjointOrbit → Functional
  ↪  → Functional → ℝ),
    (∀ ω F G, B ω F G = -B ω G F) →
    InvariantUnderCoadjointAction SDiff B →
    DegenerateWRTKineticEnergy B →
    ProducesControllableNegativeFeedback B →
    SaturatesTetherQuadratic B →
    Polarizes B →
    Polarizes TetherKernel →
    ∀ ω F G, B ω F G = ⅁_F ω F G) ∧
  (∀ (B : CoadjointOrbit → Functional → Functional → ℝ)
    (α : CoadjointOrbit → T3 → ℝ),
    HasTetherKernelDensity B α →
    (∀ ω x, α ω x = canonicalTetherDensity ω x) →
    ∀ ω F G, B ω F G = ⅁_F ω F G) ∧
  (∀ (u₀ : VelocityField) (ν : ℝ),
    0 < ν →
    (∀ x, div u₀ x = 0) →
    ContDiff ℝ (⊤ : ℕ∞) u₀ →
    Integrable (fun x : T3 => ‖u₀ x‖ ^ 2) →
    ∃ (u : ℝ → VelocityField) (p : ℝ → PressureField),
      NS_PDE u p ν ∧
      u 0 = u₀ ∧
      (∀ t ≥ (0 : ℝ), ContDiff ℝ (⊤ : ℕ∞) (u t)) ∧
      (∀ t ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
      (∀ t ≥ (0 : ℝ), ∀ x, div (u t) x = 0)) := ...
```

Blueprint variant (`ClosureBlueprint.lean`).

```
-- NS_Millennium_Proof/Modules/ClosureBlueprint.lean
public theorem conditional_regularity_of_open_hyps
    (u₀ : VelocityField) (ν : ℝ)
    (h_divfree : ∀ x, div u₀ x = 0)
    (h_smooth : ContDiff ℝ (⊤ : ℕ∞) u₀)
    (h_finite_energy : Integrable (fun x : T3 => ‖u₀ x‖ ^ 2))
    (h_pos_ν : ν > 0)
    (w : KatoLocalWitness u₀ ν)
    (hRiccati : FrohmanianTetherBound u₀ ν)
    (hcrit : TetheredLyapunov.PhysicalBKMContinuationCriterionFor u₀ ν)
    (huniq : TetheredLyapunov.KatoUniquenessOnExistenceTimes u₀ ν w) :
    ∃ (u : ℝ → VelocityField) (p : ℝ → PressureField),
      NS_PDE u p ν ∧
      u 0 = u₀ ∧
      (∀ t ≥ (0 : ℝ), ContDiff ℝ (⊤ : ℕ∞) (u t)) ∧
      (∀ t ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
      (∀ t ≥ (0 : ℝ), ∀ x, div (u t) x = 0) := ...
```

Non-vacuity and joint satisfiability at $u_0 = 0$ (`conditional_regularity_nonvacuous_at_zero`).

```
-- NS_Millennium_Proof/Modules/ClayStatements.lean
public theorem stepField_not_physical : ¬ IsPhysicalNSSolution stepField := ...
public theorem repaired_huniq_at_zero (ν : ℝ) (w : KatoLocalWitness 0 ν) : ... := ...
public theorem frohmanianTetherBound_zero (ν : ℝ) : FrohmanianTetherBound 0 ν := ...
public theorem frohmanianTetherRate_zero (ν : ℝ) : FrohmanianTetherRate 0 ν := ...

public theorem conditional_regularity_of_open_hyps_nonvacuous_at_zero (ν : ℝ) (hν : 0 < ν) :
    ∃ (u : TimeDependentVelocity),
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.IsSmooth (u t)) ∧
      (u 0 = 0) ∧
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.satisfies_NavierStokes u ν) ∧
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
      (∀ v : TimeDependentVelocity,
        (∀ t ≥ (0 : ℝ), ConditionalRegularity.IsSmooth (v t)) →
```

```
    v 0 = 0 →
    (∀ t ≥ (0 : ℝ), ConditionalRegularity.satisfies_NavierStokes v ν) →
    ∀ t ≥ (0 : ℝ), v t = u t) := ...
```

Comparison and ceiling (`IndependentMajorant.lean`, `TetheredLyapunov.lean`).

```
public theorem comparison_ode_stability
    (C κ'' : ℝ) (hC : 0 < C) (hκ : 0 < κ'')
    (y : ℝ → ℝ)
    (hdiff : ∀ s ≥ (0 : ℝ), DifferentiableAt ℝ y s)
    (hy_dot : ∀ s ≥ (0 : ℝ), deriv y s ≤ C * (y s) ^ 2 - κ'' * (y s) ^ 3)
    (t : ℝ) (ht : 0 ≤ t) :
    y t ≤ max (y 0) (C / κ'') := ...
```

```
public theorem comparison_majorant_global_bound (C κ'' y0 : ℝ) (hC : 0 < C) (hκ : 0 < κ'')
    (hy0 : 0 ≤ y0) :
    ∃ Y : ℝ, ∀ t ≥ (0 : ℝ),
      0 ≤ ComparisonODE C κ'' y0 t ∧ ComparisonODE C κ'' y0 t ≤ Y :=
  ⟨max y0 (C / κ''), fun t ht =>
    TetheredLyapunov.uniform_majorant_bound C κ'' y0 hC hκ hy0 t ht⟩
```

```
public theorem riccati_ceiling_of_vorticity_di
    (u : TimeDependentVelocity) (C κ'' : ℝ)
    (hC : 0 < C) (hκ : 0 < κ'')
    (hdiff : ∀ s ≥ (0 : ℝ),
      DifferentiableAt ℝ (fun τ => vorticity_sup_norm (vorticity (u τ))) s)
    (hDI : ∀ s ≥ (0 : ℝ),
      deriv (fun τ => vorticity_sup_norm (vorticity (u τ))) s ≤
        C * vorticity_sup_norm (vorticity (u s)) ^ 2 -
          κ'' * vorticity_sup_norm (vorticity (u s)) ^ 3)
    (hcont : Continuous fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) :
    ∃ Y : ℝ,
      (∀ τ ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u τ)) ≤ Y) ∧
      Continuous (fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) := ...
```

Conjectured pairing bridge and weighted stretching (`SymplecticTether.lean`, 0 `sorry`).

```
-- NS_Millennium_Proof/Modules/SymplecticTether.lean (sorry-free)
@[expose] public def ConjecturedUnmodifiedTetherPairing
    (u : TimeDependentVelocity) (p : TimeDependentPressure) (ν : ℝ)
    (t : ℝ) (x : T3) (C M : ℝ) : Prop :=
  0 ≤ t →
  NS_PDE u p ν →
  VorticityTransportRegularity u p t x →
  DifferentiableAt ℝ (u t) x →
  PointwiseStrainSupBound (u t) (vorticity (u t)) C x →
  inner ℝ (vorticity (u t) x) (convective (u t) (vorticity (u t)) x) = 0 →
  inner ℝ (vorticity (u t) x) (laplacian (vorticity (u t)) x) ≤ 0 →
  0 ≤ ν →
  (‖vorticity (u t) x‖ = M ∧ M = vorticity_sup_norm (vorticity (u t))) →
  ProducesControllableNegativeFeedback TetherKernel →
  kappa = CalderonZygmundConstant3D →
  inner ℝ (vorticity (u t) x) (convective (vorticity (u t)) (u t) x) ≤
    (C * M - kappa * M ^ 2) * M ^ 2

public theorem reduced_pairing_of_unmodified_and_tether
    (u : TimeDependentVelocity) (p : TimeDependentPressure) (ν : ℝ)
    (t : ℝ) (_ht : 0 ≤ t) (x : T3)
    (_hNS : NS_PDE u p ν)
    (_hreg : VorticityTransportRegularity u p t x)
    (_hu : DifferentiableAt ℝ (u t) x)
    (C M : ℝ)
    (_hCZ : ‖fderiv ℝ (u t) x‖ ≤ C * vorticity_sup_norm (vorticity (u t)))
    (_htransp : inner ℝ (vorticity (u t) x)
```

```
      (convective (u t) (vorticity (u t)) x) = 0)
    (_hvisc : inner ℝ (vorticity (u t) x)
      (laplacian (vorticity (u t)) x) ≤ 0)
    (_hν : 0 ≤ ν)
    (_hmax : ‖vorticity (u t) x‖ = M ∧
      M = vorticity_sup_norm (vorticity (u t)))
    (_htether : ProducesControllableNegativeFeedback TetherKernel)
    (_hκ : kappa = CalderonZygmundConstant3D)
    (hcancels : inner ℝ (vorticity (u t) x)
        (convective (vorticity (u t)) (u t) x) +
      kappa * M ^ 4 ≤ C * M ^ 3) :
    inner ℝ (vorticity (u t) x)
        (convective (vorticity (u t)) (u t) x) ≤
      (C * M - kappa * M ^ 2) * M ^ 2 :=
  (reduced_pairing_iff_cancels_kappa_M4
    (inner ℝ (vorticity (u t) x) (convective (vorticity (u t)) (u t) x))
    C kappa M).mpr hcancels
```

```
public theorem tethered_stretching_of_weight
    (ωField uField : VelocityField) (x : T3) (C M : ℝ)
    (hu : DifferentiableAt ℝ uField x)
    (hCZ : ‖fderiv ℝ uField x‖ ≤ C * vorticity_sup_norm ωField)
    (hmax : ‖ωField x‖ = M ∧ M = vorticity_sup_norm ωField)
    (orbit : CoadjointOrbit) (testF : Functional)
    (hweight : M ^ 4 ≤ Enstrophy orbit * ∫ y,
          ‖Pi_u (velocity_from_vorticity orbit)
              (FunctionalDerivative testF orbit) y‖ ^ 2 ∂volume) :
    inner ℝ (ωField x) (convective ωField uField x) +
        TetherKernel orbit testF testF ≤
      (C * M - kappa * M ^ 2) * M ^ 2 := ...
```

Local maximum (`NS_Equations.lean`).

```
public theorem stretching_inner_le_at_max
    (ω u : VelocityField) (x : T3) (C_CZ : ℝ)
    (hu : DifferentiableAt ℝ u x)
    (hCZ : PointwiseStrainSupBound u ω C_CZ x)
    (hmax : ‖ω x‖ = vorticity_sup_norm ω) :
    inner ℝ (ω x) (convective ω u x) ≤
      C_CZ * vorticity_sup_norm ω ^ 3 := ...
```

Tether kernel, degeneracy, invariance, symmetric uniqueness, and the conditional Jacobiator lemma (`SymplecticTether.lean`, `Uniqueness.lean`, `ForMathlib/Projection.lean`).

```
@[expose] public noncomputable def TetherKernel (ω : CoadjointOrbit) (F G : Functional) : ℝ :=
  -kappa * Enstrophy ω * ∫ x,
      inner ℝ
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative G ω) x)
    ∂volume
```

```
@[expose] public noncomputable def Pi_u (u v : VelocityField) : VelocityField :=
  fun x => v x - (
    (∫ y, pairing (v y) (u y) ∂volume) /
    (∫ y, ‖u y‖ ^ 2 ∂volume)
  ) • u x
```

```
public theorem tetherKernel_degenerates_on_kinetic_energy
    (F : Functional) (ω : CoadjointOrbit)
    (hδH : FunctionalDerivative KineticEnergyHamiltonian ω = velocity_from_vorticity ω)
    (hdiv : ∀ x, div (velocity_from_vorticity ω) x = 0)
    (hInt : Integrable (fun y => ‖velocity_from_vorticity ω y‖ ^ 2)) :
    TetherKernel ω F KineticEnergyHamiltonian = 0 := ...
```

```
-- NS_Millennium_Proof/Modules/ArnoldGeometric.lean (abstract SDiff action; replaces the former
↪  identity stub)
public abbrev CoadjointAction {SDiff : Type*} [Group SDiff] [MulAction SDiff CoadjointOrbit]
    (g : SDiff) (ω : CoadjointOrbit) : CoadjointOrbit :=
  g • ω

public structure SDiffActionHyp (SDiff : Type*) [Group SDiff] [MulAction SDiff CoadjointOrbit] :
    Prop where
  casimir_invariant : ∀ (f : ℝ → ℝ) (g : SDiff) (ω : CoadjointOrbit),
    (∫ x, f ‖(g • ω).val x‖ ∂volume) = ∫ x, f ‖ω.val x‖ ∂volume
  transitive_on_casimir_levels : ∀ ω₁ ω : CoadjointOrbit,
    (∀ f : ℝ → ℝ, Casimir f ω₁ = Casimir f ω) → ∃ g : SDiff, g • ω₁ = ω
```

```
-- NS_Millennium_Proof/Modules/SymplecticTether.lean
@[expose] public def TetherIntegralPreserved (SDiff : Type*) [Group SDiff]
    [MulAction SDiff CoadjointOrbit] : Prop :=
  ∀ (g : SDiff) (F G : Functional) (ω : CoadjointOrbit),
    (∫ x,
      inner ℝ
        (Pi_u (velocity_from_vorticity (CoadjointAction g ω))
          (FunctionalDerivative F (CoadjointAction g ω)) x)
        (Pi_u (velocity_from_vorticity (CoadjointAction g ω))
          (FunctionalDerivative G (CoadjointAction g ω)) x) ∂volume) =
    ∫ x,
      inner ℝ
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative G ω) x) ∂volume

public theorem tether_coadjoint_invariance (SDiff : Type*) [Group SDiff]
    [MulAction SDiff CoadjointOrbit] (hyp : SDiffActionHyp SDiff)
    (h_density : TetherIntegralPreserved SDiff)
    (g : SDiff) (F G : Functional) (ω : CoadjointOrbit) :
    TetherKernel (CoadjointAction g ω) F G = TetherKernel ω F G := by
  unfold TetherKernel
  rw [hyp.enstrophy_invariant g ω, h_density g F G ω]
```

```
public def SaturatesTetherQuadratic
    (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop :=
  ∀ (F : Functional) (ω : CoadjointOrbit), B ω F F = TetherKernel ω F F
```

```
-- NS_Millennium_Proof/Modules/Uniqueness.lean
public theorem step2_quadratic_and_locality
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_sym : ∀ ω F G, B ω F G = B ω G F)
    (h_proj : ∀ ω (F F' G : Functional),
      Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) =
        Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F' ω) → B ω F G = B ω F' G) :
    FactorsThroughProjection B ∧
    ∀ ω (F G F' G' : Functional), FunctionalDerivative F ω = FunctionalDerivative F' ω →
      FunctionalDerivative G ω = FunctionalDerivative G' ω → B ω F G = B ω F' G' :=
  have h2 := step2_quadratic B h_sym h_proj
  ⟨h2, step1_of_factorization B h2⟩

public theorem uniqueness_of_minimal_tether_core
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_sym : ∀ ω F G, B ω F G = B ω G F)
    (h_proj : ∀ ω (F F' G : Functional),
      Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) =
        Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F' ω) → B ω F G = B ω F' G)
    (h_density : FactorsThroughProjection B → ∃ α, HasTetherKernelDensity B α)
    (h_weight : ∀ α, HasTetherKernelDensity B α → ∃ c : ℝ, ∀ ω x, α ω x = -c * Enstrophy ω)
    (h_sat : SaturatesTetherQuadratic B)
    (h_nondeg : ∃ (ω : CoadjointOrbit) (F : Functional), Enstrophy ω * (∫ x,
        inner ℝ (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
          (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x) ∂NavierStokes3D.volume)  0)
          ↪  :
    ∀ ω F G, B ω F G = TetherKernel ω F G :=
  have h2 := (step2_quadratic_and_locality B h_sym h_proj).1
```

```
  have h3 := step3_scalar B h2 h_density h_weight
  have h4 := step4_scalar_eq_kappa B h3 h_sat h_nondeg
  step5_eq_tether B h4

public theorem uniqueness_of_minimal_tether
    (SDiff : Type*) [Group SDiff] [MulAction SDiff CoadjointOrbit]
    (_hyp : SDiffActionHyp SDiff)
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_sym : ∀ ω F G, B ω F G = B ω G F)
    (hC1 : InvariantUnderCoadjointAction SDiff B)
    (hC2 : DegenerateWRTKineticEnergy B)
    (hC3 : ProducesControllableNegativeFeedback B)
    (h_sat : SaturatesTetherQuadratic B)
    (h_polarB : Polarizes B)
    (h_polarT : Polarizes TetherKernel) :
    ∀ (ω : CoadjointOrbit) (F G : Functional), B ω F G = TetherKernel ω F G :=
  uniqueness_of_minimal_symmetric_tether SDiff B h_sym hC1 hC2 hC3 h_sat h_polarB h_polarT
```

```
public theorem tethered_jacobi_identity (F G H : Functional) (ω : CoadjointOrbit)
    (X Y Z : VelocityField)
    (_hF : FunctionalDerivative F ω = X)
    (_hG : FunctionalDerivative G ω = Y)
    (_hH : FunctionalDerivative H ω = Z)
    (hident : jacobiator F G H ω =
      ∫ x, cyclicLiePairing X Y Z x ∂volume)
    (hvanish : (∫ x, cyclicLiePairing X Y Z x ∂volume) = 0) :
    jacobiator F G H ω = 0 := ...
```

Lyapunov vehicle (`TetheredLyapunov.lean`).

```
public def IsEnstrophyAccumulation (u : ℝ → VelocityField) (ω : ℝ → VorticityField)
    (φ : ℝ → T3 → ℝ) : Prop :=
  ∀ t ≥ (0 : ℝ), ∀ x, MaterialDerivative u φ t x = ‖ω t x‖ ^ 2
```

```
public noncomputable def LyapunovS (ωε : VorticityField) (phi : T3 → ℝ) : ℝ :=
  ∫ x, (1 / 2) * ‖ωε x‖ ^ 2 + (kappa / 4) * ‖ωε x‖ ^ 4 * phi x ∂volume
```

Frohmanian tether bound (`SymplecticTether.lean`).

```
-- NS_Millennium_Proof/Modules/SymplecticTether.lean
@[expose] public abbrev FrohmanianTetherBound (u₀ : VelocityField) (ν : ℝ) : Prop :=
  ∀ u : ℝ → VelocityField, ∀ p : ℝ → PressureField,
    NS_PDE u p ν → u 0 = u₀ → IsPhysicalNSSolution u →
      ∃ Y : ℝ,
        (∀ τ ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u τ)) ≤ Y) ∧
        Continuous (fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) ∧
        (∀ τ ≥ (0 : ℝ), BddAbove (Set.range fun x => ‖vorticity (u τ) x‖))
```

Historical audit of unpatched binder (`ClayStatements.lean`).

```
public theorem library_huniq_unsatisfiable_at_zero (ν : ℝ) (w : KatoLocalWitness 0 ν) :
    ¬ (∀ (T' : ℝ) (v : TimeDependentVelocity) (q : TimeDependentPressure),
      T' ∈ existenceTimes 0 ν → v 0 = 0 → NS_PDE v q ν →
      (∀ t ∈ Set.Icc (0 : ℝ) T', ContDiff ℝ (⊤ : ℕ∞) (v t)) →
      ∀ t ∈ Set.Icc (0 : ℝ) T', w.u t = v t) := ...
```

```
public theorem closing_bound_uniqueness_family_unsatisfiable (ν : ℝ)
    (hKato : ∀ u₀ : VelocityField, ContDiff ℝ (⊤ : ℕ∞) u₀ → (∀ x, div u₀ x = 0) →
      Integrable (fun x : T3 => ‖u₀ x‖ ^ 2) → KatoLocalWitness u₀ ν) :
    ¬ (∀ (u₀ : VelocityField) (w : KatoLocalWitness u₀ ν)
      (T' : ℝ) (v : TimeDependentVelocity) (q : TimeDependentPressure),
      T' ∈ existenceTimes u₀ ν → v 0 = u₀ → NS_PDE v q ν →
      (∀ t ∈ Set.Icc (0 : ℝ) T', ContDiff ℝ (⊤ : ℕ∞) (v t)) →
      ∀ t ∈ Set.Icc (0 : ℝ) T', w.u t = v t) := ...
```

Periodic Clay (B) hypotheses (`ClayStatements.lean`).

```
@[expose] public def PeriodicTetherClosingBound (ν : ℝ) : Prop :=
  ∀ u₀ : VelocityField,
    (∀ x, div u₀ x = 0) → ContDiff ℝ (⊤ : ℕ∞) u₀ → IsZ3Periodic u₀ →
      ∃ Y : ℝ, ∀ (T : ℝ) (u : TimeDependentVelocity) (p : TimeDependentPressure),
        IsPeriodicSmoothSolutionOn ν T u₀ u p →
          ∀ t ∈ Set.Ico (0 : ℝ) T, vorticity_sup_norm (vorticity (u t)) ≤ Y
```

```
@[expose] public def PeriodicBKMContinuation (ν : ℝ) : Prop :=
  ∀ u₀ : VelocityField,
    (∀ x, div u₀ x = 0) → ContDiff ℝ (⊤ : ℕ∞) u₀ → IsZ3Periodic u₀ →
      (∃ Y : ℝ, ∀ (T : ℝ) (u : TimeDependentVelocity) (p : TimeDependentPressure),
        IsPeriodicSmoothSolutionOn ν T u₀ u p →
          ∀ t ∈ Set.Ico (0 : ℝ) T, vorticity_sup_norm (vorticity (u t)) ≤ Y) →
        ∃ (u : TimeDependentVelocity) (p : TimeDependentPressure),
          IsPeriodicSmoothSolution ν u₀ (fun _ _ => 0) u p
```

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